W6_Class_Project_ATORBATI

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```r
#install.packages("tinytex")
#tinytex::install_tinytex()
#install.packages("viridis")
```

```r
library(tidyverse)
library(knitr)
library(viridis)
```

```r
# ---- Load data ----
file_path <- "data/NEWDATASET_VER1.csv"
stopifnot(file.exists(file_path))

wonder_raw <- readr::read_csv(file_path, show_col_types = FALSE)
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```r
#Clean names - I had to use chat gpt to help me with name cleaning
clean_names_simple <- function(nms) {
  nms <- gsub("[^A-Za-z0-9]+", "_", nms)
  nms <- gsub("^_+|_+$", "", nms)
  tolower(nms)
}
names(wonder_raw) <- clean_names_simple(names(wonder_raw))

# Standardize types + reliability flag
#I also used chatgpt for help on this part as I was getting odd results without it.
wonder <- wonder_raw %>%
  mutate(
    state = as.factor(state),
    year  = as.integer(year),
    sex   = as.factor(sex),
    race  = as.factor(race),
    deaths = as.numeric(deaths),
    population = as.numeric(population),
    crude_rate_chr = as.character(crude_rate),
    crude_rate = suppressWarnings(as.numeric(crude_rate_chr)),
    unreliable_rate = (tolower(crude_rate_chr) == "unreliable") | (deaths < 20)
```

```r
  )

# Confirm key columns exist
c("state","year","sex","race","deaths","population","crude_rate") %in% names(wonder)
```

```
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

```r
# Main analysis tables
state_year <- wonder %>%
  group_by(state, year) %>%
  summarise(
    deaths = sum(deaths, na.rm = TRUE),
    population = sum(population, na.rm = TRUE),
    rate_per_100k = 100000 * deaths / population,
    .groups = "drop"
  )

state_year_sex <- wonder %>%
  group_by(state, year, sex) %>%
  summarise(
    deaths = sum(deaths, na.rm = TRUE),
    population = sum(population, na.rm = TRUE),
    rate_per_100k = 100000 * deaths / population,
    .groups = "drop"
  ) %>%
  filter(sex %in% c("Male", "Female"))

state_year_race <- wonder %>%
  group_by(state, year, race) %>%
  summarise(
    deaths = sum(deaths, na.rm = TRUE),
    population = sum(population, na.rm = TRUE),
    rate_per_100k = 100000 * deaths / population,
    .groups = "drop"
  )
```

Three different statistical tests

1) t-test (Sex) [Type: mean comparison] -> RQ3

2) ANOVA (Race) [Type: mean comparison] -> RQ3

3) OLS Regression [Type: regression] -> RQ1/RQ2

Each includes: results + analysis + conclusion tied to RQs

```r
# TEST 1 (RQ3): Two-sample t-test comparing mean rates by Sex

# Descriptives
sex_means <- state_year_sex %>%
  group_by(sex) %>%
  summarise(
    mean_rate = mean(rate_per_100k, na.rm = TRUE),
    sd_rate   = sd(rate_per_100k, na.rm = TRUE),
    n         = n(),
    .groups = "drop"
  )
kable(sex_means, digits = 2, caption = "Descriptives: state-year suicide rates by sex")
```

Table 1: Descriptives: state-year suicide rates by sex

sex	mean_rate	sd_rate	n
Female	1.64	2.65	924
Male	4.38	2.50	1102

```r
# Test
t_sex <- t.test(rate_per_100k ~ sex, data = state_year_sex)
print(t_sex)
```

```
Welch Two Sample t-test
```

data: rate_per_100k by sex t = -23.743, df = 1918.8, p-value < 2.2e-16 alternative hypothesis: true difference in means between group Female and group Male is not equal to 0 95 percent confidence interval: -2.961724 -2.509772 sample estimates: mean in group Female mean in group Male 1.639507 4.375255

```r
# Analysis + conclusion
cat("\n**Analysis (t-test, RQ3):** This two-sample t-test evaluates whether the mean state-year suicide
```

Analysis (t-test, RQ3): This two-sample t-test evaluates whether the mean state-year suicide mortality rate differs between males and females. A statistically significant p-value indicates evidence of a difference in average rates by sex in this dataset.

```r
if (t_sex$p.value < 0.05) {
  cat("**Conclusion (t-test, RQ3):** The test is statistically significant (p < 0.05), suggesting mean s
} else {
  cat("**Conclusion (t-test, RQ3):** The test is not statistically significant (p >= 0.05), suggesting r
}
```

Conclusion (t-test, RQ3): The test is statistically significant ($p < 0.05$), suggesting mean suicide mortality rates differ by sex. This is consistent with expectations that male rates are higher on average.

```r
cat("**Expectation check:** If results differ from expectations, likely causes include aggregation level
```

Expectation check: If results differ from expectations, likely causes include aggregation level (state-year), differences in population composition by state, and the need for models that control for state and year simultaneously.

```r
# TEST 2 (RQ3): One-way ANOVA comparing mean rates across Race categories

# Descriptives
race_means <- state_year_race %>%
  group_by(race) %>%
  summarise(
    mean_rate = mean(rate_per_100k, na.rm = TRUE),
    sd_rate   = sd(rate_per_100k, na.rm = TRUE),
    n         = n(),
    .groups = "drop"
  ) %>%
  arrange(desc(mean_rate))
kable(race_means, digits = 2, caption = "Descriptives: state-year suicide rates by race")
```

Table 2: Descriptives: state-year suicide rates by race

race	mean_rate	sd_rate	n
American Indian or Alaska Native	19.73	12.48	125
Asian or Pacific Islander	4.15	2.48	160
Black or African American	3.96	1.98	528
White	3.61	2.33	1101
NA	NaN	NA	1

```r
# ANOVA
aov_race <- aov(rate_per_100k ~ race, data = state_year_race)
aov_summary <- summary(aov_race)
print(aov_summary)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)
```

race 3 29863 9954 671.3 <2e-16 *** Residuals 1910 28321 15
— Signif. codes: 0 ‘’ ***0.001*** ’’ *0.01* ” 0.05 ‘.’ 0.1 ’ ’ 1 1 observation deleted due to missingness

```r
cat("\n**Analysis (ANOVA, RQ3):** One-way ANOVA tests whether mean state-year suicide rates differ acros
```

Analysis (ANOVA, RQ3): One-way ANOVA tests whether mean state-year suicide rates differ across race categories. A significant F-test indicates that at least one race group mean differs from the others.

```r
p_race <- aov_summary[[1]][["Pr(>F)"]][1]
if (!is.na(p_race) && p_race < 0.05) {
  cat("**Conclusion (ANOVA, RQ3):** The F-test is statistically significant (p < 0.05), providing evide
} else {
  cat("**Conclusion (ANOVA, RQ3):** The F-test is not statistically significant (p >= 0.05), suggesting
}
```

Conclusion (ANOVA, RQ3): The F-test is statistically significant ($p < 0.05$), providing evidence that mean suicide mortality rates differ across race categories. A post-hoc test (Tukey HSD) can identify which pairs differ.

```r
cat("**Expectation check:** If results are weaker than expected, potential reasons include aggregation e
```

Expectation check: If results are weaker than expected, potential reasons include aggregation effects, differences in state-level demographics, and limited within-category variation after collapsing across covariates.

```r
# OPTIONAL follow-up if ANOVA is significant:
# print(TukeyHSD(aov_race))
```

```r
# TEST 3 (RQ1/RQ2): OLS regression with state fixed effects + linear time trend
# DV: rate_per_100k
# IVs: year (time trend), state (fixed effects)

reg1 <- lm(rate_per_100k ~ year + state, data = state_year)
reg1_sum <- summary(reg1)
print(reg1_sum)
```

Call: lm(formula = rate_per_100k ~ year + state, data = state_year)

Residuals: Min 1Q Median 3Q Max -4.6389 -0.2539 -0.0253 0.2021 10.5333

Coefficients: Estimate Std. Error t value Pr(>|t|)
(Intercept) -71.583603 8.827495 -8.109 1.41e-15 ***year 0.037442 0.004392 8.526 < 2e-16*** stateAlaska 9.098643 0.279082 32.602 < 2e-16 ***stateArizona -1.126949 0.279082 -4.038 5.78e-05*** stateArkansas 0.620007 0.279082 2.222 0.026523 *
stateCalifornia -2.761307 0.279082 -9.894 < 2e-16 ***stateColorado -1.088981 0.279082 -3.902 0.000101*** stateConnecticut -1.221322 0.279082 -4.376 1.33e-05 ***stateDelaware 3.464915 0.279082 12.415 < 2e-16*** stateDistrict of Columbia 4.146836 0.683608 6.066 1.83e-09 ***stateFlorida -2.156212 0.279082 -7.726 2.58e-14*** stateGeorgia -1.165512 0.279082 -4.176 3.21e-05 ***stateHawaii 3.068854 0.279082 10.996 < 2e-16*** stateIdaho 1.016712 0.279082 3.643 0.000283 ***stateIllinois -2.400414 0.279082 -8.601 < 2e-16*** stateIndiana -1.110190 0.279082 -3.978 7.43e-05 ***stateIowa -0.898806 0.279082 -3.221 0.001319*** *stateKansas -0.364606 0.279082 -1.306 0.191686*
stateKentucky -0.347776 0.279082 -1.246 0.212989
stateLouisiana 0.520573 0.279082 1.865 0.062417 .
stateMaine 1.030197 0.279082 3.691 0.000235 **stateMaryland -1.218664 0.279082 -4.367 1.39e-05**
stateMassachusetts -2.209297 0.279082 -7.916 6.19e-15 **stateMichigan -1.743699 0.279082 -6.248 6.03e-10** *stateMinnesota -1.684221 0.279082 -6.035 2.20e-09* **stateMississippi 0.667902 0.279082 2.393 0.016876 ***
stateMissouri -1.238736 0.279082 -4.439 1.00e-05 *stateMontana 3.191918 0.279082 11.437 < 2e-16*
stateNebraska 0.558874 0.279082 2.003 0.045483 *
stateNevada 0.226483 0.279082 0.812 0.417246
stateNew Hampshire 0.484512 0.279082 1.736 0.082841 .
stateNew Jersey -2.260784 0.279082 -8.101 1.51e-15 *stateNew Mexico 1.335816 0.279082 4.786 1.94e-06* **stateNew York -2.646614 0.279082 -9.483 < 2e-16** *stateNorth Carolina -1.497218 0.279082 -5.365 9.97e-08* **stateNorth Dakota 2.178654 0.279082 7.807 1.42e-14** *stateOhio -1.967114 0.279082 -7.049 3.27e-12* **stateOklahoma 0.264236 0.279082 0.947 0.343955**
stateOregon -0.927321 0.279082 -3.323 0.000922 *statePennsylvania -1.912925 0.279082 -6.854 1.22e-11* **stateRhode Island 1.314622 0.279082 4.711 2.80e-06** *stateSouth Carolina 0.108971 0.279082 0.390 0.696274*
stateSouth Dakota 4.241419 0.279082 15.198 < 2e-16 **stateTennessee -0.674690 0.279082 -2.418 0.015796 ***
stateTexas -2.394089 0.279082 -8.578 < 2e-16 *stateUtah -0.178538 0.279082 -0.640 0.522486*
stateVermont 3.602682 0.279082 12.909 < 2e-16 **stateVirginia -1.591705 0.279082 -5.703 1.53e-08** *stateWashington -1.711051 0.279082 -6.131 1.23e-09* **stateWest Virginia 0.817392 0.279082**

2.929 0.003476 stateWisconsin -1.532997 0.279082 -5.493 4.96e-08 ***stateWyoming 5.208161 0.279082 18.662 < 2e-16*** — Signif. codes: 0 '*** 0.001 ** 0.01* ** 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9256 on 1050 degrees of freedom (1 observation deleted due to missingness) Multiple R-squared: 0.8667, Adjusted R-squared: 0.8602 F-statistic: 133.9 on 51 and 1050 DF, p-value: < 2.2e-16

```
beta_year <- coef(reg1_sum)["year", "Estimate"]
p_year <- coef(reg1_sum)["year", "Pr(>|t|)"]

cat("\n**Analysis (OLS regression, RQ1/RQ2):** This model estimates the association between time (year)
```

Analysis (OLS regression, RQ1/RQ2): This model estimates the association between time (year) and the suicide rate while controlling for persistent differences across states via state fixed effects. The coefficient on year is the estimated average annual change in deaths per 100,000.

```
if (!is.na(p_year) && p_year < 0.05) {
  cat(paste0("**Conclusion (OLS regression, RQ1/RQ2):** The year coefficient is statistically significa
} else {
  cat("**Conclusion (OLS regression, RQ1/RQ2):** The year coefficient is not statistically significant
}
```

Conclusion (OLS regression, RQ1/RQ2): The year coefficient is statistically significant ($p < 0.05$). This supports a time trend in suicide mortality rates. Estimated annual change approximately 0.0374 deaths per 100,000 per year.

```
cat("**Expectation check:** If the trend differs from expectations, reasons may include nonlinear chang
```

Expectation check: If the trend differs from expectations, reasons may include nonlinear changes over time, differing state-specific trends, or changes in the underlying population denominator.

W6 Assignment

1) One additional figure + analysis (NOT similar to existing plots)

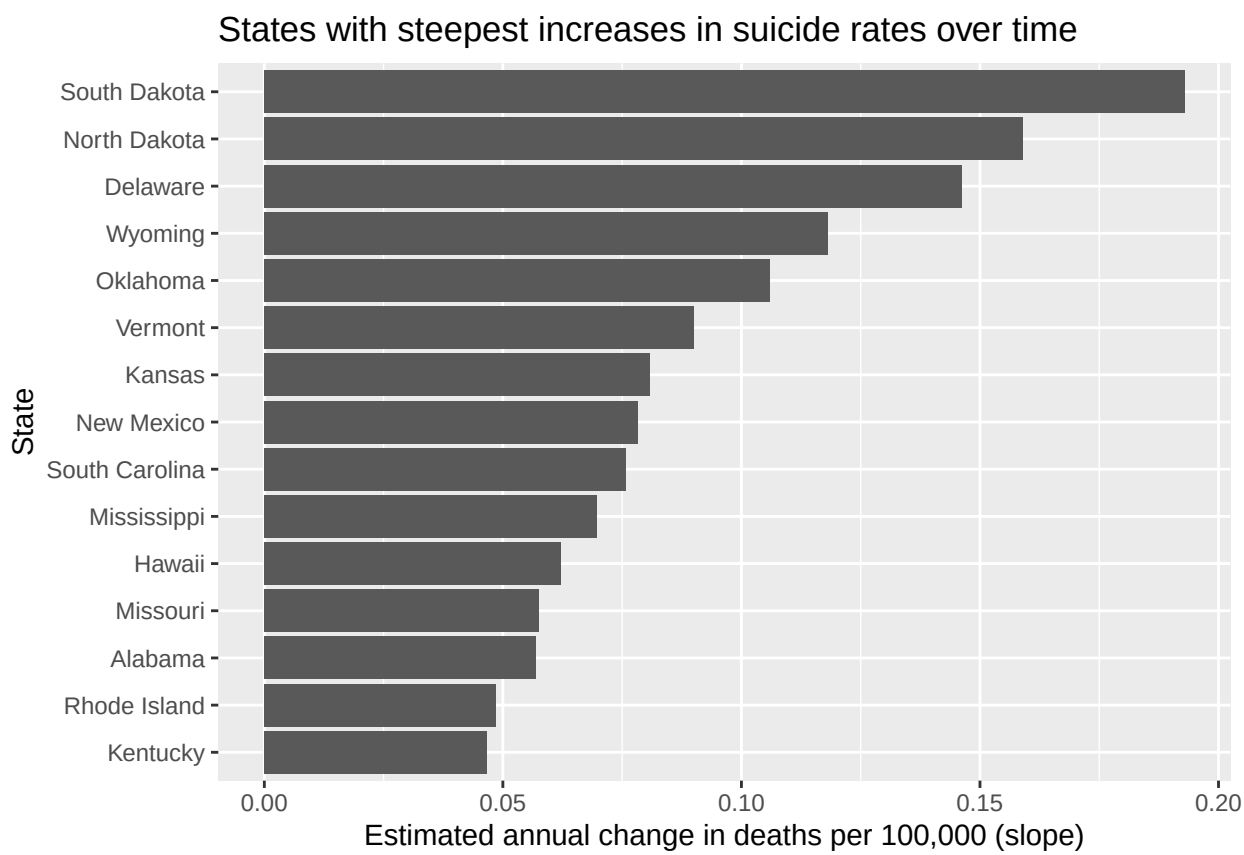
2) One additional statistical test + analysis (NOT similar to prior tests)

```
# ADDITIONAL FIGURE (NEW): State-specific slopes (who is increasing fastest?)
# Not similar to: national trend line, top-state trend lines, heatmap, boxplots

state_slopes <- state_year %>%
  group_by(state) %>%
  filter(n() >= 5) %>%  # avoid short series
  summarise(
    slope = coef(lm(rate_per_100k ~ year))[2],
    avg_rate = mean(rate_per_100k, na.rm = TRUE),
    .groups = "drop"
```

```r
  ) %>%
  arrange(desc(slope)) %>%
  slice_head(n = 15)

ggplot(state_slopes, aes(x = reorder(state, slope), y = slope)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "States with steepest increases in suicide rates over time",
    x = "State",
    y = "Estimated annual change in deaths per 100,000 (slope)"
  )
```




```r
# Analysis to include in narrative:
# This figure estimates a separate linear trend within each state and ranks states by slope.
# States with larger positive slopes are experiencing faster increases over time, which helps
# identify where policy attention may be most needed (supports RQ2).
```

```r
# ADDITIONAL STATISTICAL TEST (NEW): Poisson regression with population offset
# Not similar to OLS/t-test/ANOVA because:
# - Outcome is death counts (deaths)
# - Uses Poisson distribution and log link
# - Offset log(population) models rates using a count framework
#
# RQ1/RQ2: time trend + state differences in a count model
```

```r
poisson_fit <- glm(
  deaths ~ year + state,
  family = poisson(link = "log"),
  offset = log(population),
  data = state_year
)

poisson_sum <- summary(poisson_fit)
print(poisson_sum)
```

Call: glm(formula = deaths ~ year + state, family = poisson(link = "log"), data = state_year, offset = log(population))

Coefficients: Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.547e+01 3.774e-01 -93.972 < 2e-16 ***year 1.256e-02 1.877e-04 66.936 < 2e-16*** stateAlaska 1.133e+00 2.352e-02 48.177 < 2e-16 ***stateArizona -3.783e-01 1.117e-02 -33.866 < 2e-16*** stateArkansas 1.516e-01 1.395e-02 10.863 < 2e-16 ***stateCalifornia -1.405e+00 9.520e-03 -147.616 < 2e-16*** stateColorado -3.545e-01 1.153e-02 -30.755 < 2e-16 ***stateConnecticut -4.139e-01 1.622e-02 -25.516 < 2e-16*** stateDelaware 6.389e-01 3.035e-02 21.052 < 2e-16 ***stateDistrict of Columbia 7.551e-01 2.002e-01 3.772 0.000162*** stateFlorida -8.910e-01 9.755e-03 -91.331 < 2e-16 ***stateGeorgia -3.876e-01 1.107e-02 -35.006 < 2e-16*** stateHawaii 5.878e-01 2.377e-02 24.734 < 2e-16 ***stateIdaho 2.075e-01 1.669e-02 12.428 < 2e-16*** stateIllinois -1.067e+00 1.094e-02 -97.534 < 2e-16 ***stateIndiana -3.589e-01 1.174e-02 -30.580 < 2e-16*** stateIowa -2.939e-01 1.482e-02 -19.825 < 2e-16 ***stateKansas -1.055e-01 1.457e-02 -7.242 4.43e-13*** stateKentucky -1.068e-01 1.254e-02 -8.513 < 2e-16 ***stateLouisiana 1.290e-01 1.303e-02 9.905 < 2e-16*** stateMaine 2.325e-01 1.981e-02 11.735 < 2e-16 ***stateMaryland -4.143e-01 1.361e-02 -30.435 < 2e-16*** stateMassachusetts -9.324e-01 1.351e-02 -69.021 < 2e-16 ***stateMichigan -6.546e-01 1.093e-02 -59.916 < 2e-16*** stateMinnesota -6.230e-01 1.300e-02 -47.926 < 2e-16 ***stateMississippi 1.595e-01 1.491e-02 10.699 < 2e-16*** stateMissouri -4.109e-01 1.163e-02 -35.329 < 2e-16 ***stateMontana 5.957e-01 1.902e-02 31.315 < 2e-16*** stateNebraska 1.116e-01 1.921e-02 5.812 6.18e-09 ***stateNevada 4.322e-02 1.368e-02 3.160 0.001576*** *stateNew Hampshire 1.004e-01 2.036e-02 4.930 8.20e-07* **stateNew Jersey -9.637e-01 1.276e-02 -75.516 < 2e-16** *stateNew Mexico 3.034e-01 1.470e-02 20.633 < 2e-16* **stateNew York -1.288e+00 1.061e-02 -121.320 < 2e-16** *stateNorth Carolina -5.352e-01 1.098e-02 -48.744 < 2e-16* **stateNorth Dakota 4.667e-01 2.812e-02 16.594 < 2e-16** *stateOhio -7.726e-01 1.061e-02 -72.782 < 2e-16* **stateOklahoma 5.692e-02 1.286e-02 4.425 9.65e-06** *stateOregon -3.006e-01 1.259e-02 -23.876 < 2e-16* **statePennsylvania -7.434e-01 1.042e-02 -71.356 < 2e-16** *stateRhode Island 2.745e-01 3.131e-02 8.768 < 2e-16* **stateSouth Carolina 2.760e-02 1.274e-02 2.167 0.030201 * stateSouth Dakota 7.693e-01 2.429e-02 31.670 < 2e-16** *stateTennessee -2.143e-01 1.148e-02 -18.660 < 2e-16* **stateTexas -1.066e+00 9.720e-03 -109.622 < 2e-16** *stateUtah -5.755e-02 1.392e-02 -4.135 3.55e-05* **stateVermont 6.779e-01 2.902e-02 23.359 < 2e-16** *stateVirginia -5.755e-01 1.144e-02 -50.301 < 2e-16* **stateWashington -6.357e-01 1.150e-02 -55.285 < 2e-16** *stateWest Virginia 1.834e-01 1.611e-02 11.382 < 2e-16* **stateWisconsin -5.442e-01 1.223e-02 -44.512 < 2e-16** *stateWyoming 8.528e-01 2.506e-02 34.030 < 2e-16* ** — Signif. codes: 0 ‘*** 0.001 ** 0.01 * 0.05 ‘.’ 0.1 ’ ’ 1

(Dispersion parameter for poisson family taken to be 1)

```
Null deviance: 187490.9  on 1101  degrees of freedom
```

Residual deviance: 6033.5 on 1050 degrees of freedom (1 observation deleted due to missingness) AIC: 14764

Number of Fisher Scoring iterations: 4

```r
beta_year_pois <- coef(poisson_sum)["year", "Estimate"]
p_year_pois <- coef(poisson_sum)["year", "Pr(>|z|)"]
irr <- exp(beta_year_pois)

cat("\n**Analysis (Poisson regression, RQ1/RQ2):** Poisson regression models death counts while account
```

Analysis (Poisson regression, RQ1/RQ2): Poisson regression models death counts while accounting for population exposure using an offset. exp(beta_year) is the incidence rate ratio (IRR), interpreted as the multiplicative change in expected deaths per year (holding state constant and adjusting for population).

```r
if (!is.na(p_year_pois) && p_year_pois < 0.05) {
  cat(paste0("**Conclusion (Poisson regression, RQ1/RQ2):** The year effect is statistically significan
             ", meaning expected deaths increase by about ", round((irr - 1) * 100, 2), "% per year (sta
} else {
  cat("**Conclusion (Poisson regression, RQ1/RQ2):** The year effect is not statistically significant (
}
```

Conclusion (Poisson regression, RQ1/RQ2): The year effect is statistically significant ($p < 0.05$). IRR approximately 1.0126, meaning expected deaths increase by about 1.26% per year (state fixed, population-adjusted).

```r
cat("**Model choice justification:** If overdispersion is present (variance >> mean), a Negative Binomia
```

Model choice justification: If overdispersion is present (variance » mean), a Negative Binomial model is more appropriate than Poisson. I will check dispersion diagnostics before selecting Poisson vs Negative Binomial for the final model.